1. Write a C# Sharp program to compute the sum of the two numerical values. If the two values are the same, return triple their sum.
Sample Input:
1, 2
3, 2
2, 2
Expected Output:
3
5
12

2. Write a C# Sharp program to get the absolute difference between n and 51. If n is broader than 51 return triple the absolute difference.
Sample Input:
53
30
51
Expected Output:
6
21
0

3. Write a C# Sharp program to check two given integers, and return true if one of them is 30 or if their sum is 30.
Sample Input:
30, 0
25, 5
20, 30
20, 25
Expected Output:
True
True
True
False

4. Write a C# Sharp program to check a given integer and return true if it is within 10 of 100 or 200.
Sample Input:
103

90
89
Expected Output:
True
True
False

5. Write a C# Sharp program to create a string where 'if' is added to the front of a given string. If the string already begins with 'if', return it unchanged.
Sample Input:
"if else"
"else"
Expected Output:
if else
if else

6. Write a C# Sharp program to remove the character at a given position in the string. The given position will be in the range 0..(string length -1) inclusive.
Sample Input:
"Python", 1
"Python", o
"Python", 4
Expected Output:
Pthon
ython
Pythn

7. Write a C# Sharp program to exchange the first and last characters in a given string and return the new string.
Sample Input:
"abcd"
"a"
"xy"
Expected Output:
dbca
a
yx

8. Write a C# Sharp program to create a string which is 4 copies of the 2 front characters of a given string. If the given string length is less than 2 return the original string.
Sample Input:
"C Sharp"
"JS"
"a"
Expected Output:
C C C C
JSJSJSJS
a

9. Write a C# Sharp program to create a string with the last char added at the front and back of a given string of length 1 or more.
Sample Input:
"Red"
"Green"
"1"
Expected Output:
dRedd
nGreenn
111

10. Write a C# Sharp program to check if a given positive number is a multiple of 3 or 7.
Sample Input:
3
14
12
37
Expected Output:
True
True
True
False

11. Write a C# Sharp program to create a string taking the first 3 characters of a given string. Return the string with the 3 characters added at both the front and back. If the given string length is less than 3, use whatever characters are there.

Sample Input:
"Python"
"JS"
"Code"
Expected Output:
PytPythonPyt
JSJSJS
CodCodeCod

12. Write a C# Sharp program to check if a given string starts with 'C#' or not.
Sample Input:
"C# Sharp"
"C#"
"C++"
Expected Output:
True
True
False

13. Write a C# Sharp program that checks whether one temperature is less than 0 and another is greater than 100.
Sample Input:
120, -1
-1, 120
2, 120
Expected Output:
True
True
False

14. Write a C# Sharp program to check two given integers whether either of them is in the range 100..200 inclusive.
Sample Input:
100, 199
250, 300
105, 190
Expected Output:
True
False
True

15. Write a C# Sharp program to check whether three given integer values are in the range 20..50 inclusive. Return true if 1 or more of them are in the said range otherwise false.
Sample Input:
11, 20, 12
30, 30, 17
25, 35, 50
15, 12, 8
Expected Output:
True
True
True
False

16. Write a C# Sharp program to check whether two given integer values are in the range 20..50 inclusive. Return true if 1 or other is in the range, otherwise false.
Sample Input:
20, 84
14, 50
11, 45
25, 40
Expected Output:
True
True
True
False

17. Write a C# Sharp program to check if a string 'yt' appears at index 1 in a given string. If it appears return a string without 'yt' otherwise return the original string.
Sample Input:
"Python"
"ytade"
"jsues"
Expected Output:
Phon
ytade
jsues

18. Write a C# Sharp program to check the largest number among three given integers.
Sample Input:
1,2,3
1,3,2
1,1,1
1,2,2
Expected Output:
3
3
1
2

19. Write a C# Sharp program to check which number is closest to 100 among two given integers. Return 0 if the two numbers are equal.
Sample Input:
78, 95
95, 95
99, 70
Expected Output:
95
0
99

20. Write a C# Sharp program to check whether two given integers are in the range 40..50 inclusive, or they are both in the range 50..60 inclusive.
Sample Input:
78, 95
25, 35
40, 50
55, 60
Expected Output:
False
False
True
True

21. Write a C# Sharp program to find the largest value from two positive integer values in the range 20..30 inclusive. Return 0 if neither is in that range.

Sample Input:
78, 95
20, 30
21, 25
28, 28
Expected Output:
0
30
25
28

22. Write a C# Sharp program to check if a given string contains between 2 and 4 'z' characters.
Sample Input:
"frizz"
"zane"
"Zazz"
"false"
"zzzz"
"ZZZZ"
Expected Output:
True
False
True
False
True
False

23. Write a C# Sharp program to check if two given non-negative integers have the same last digit.
Sample Input:
123, 456
12, 512
7, 87
12, 45
Expected Output:
False
True
True

False

24. Write a C# Sharp program to convert the last 3 characters of a given string to uppercase. If the length of the string is less than 3, then uppercase all the characters.
Sample Input:
"Python"
"Javascript"
"js"
"PHP"
Expected Output:
PytHON
JavascrIPT
JS
PHP

25. Write a C# Sharp program to create a string which is n (non-negative integer) copies of a given string.
Sample Input:
"JS", 2
"JS", 3
"JS", 1
Expected Output:
JSJS
JSJSJS
JS

1. Write a C# Sharp program to accept two integers and check whether they are equal or not.
Test Data :
Input 1st number: 5
Input 2nd number: 5
Expected Output :
5 and 5 are equal

2. Write a C# Sharp program to check whether a given number is even or odd.
Test Data : 15
Expected Output :
15 is an odd integer

3. Write a C# Sharp program to check whether a given number is positive or negative.
Test Data : 14
Expected Output :
14 is a positive number

4. Write a C# Sharp program to find out whether a given year is a leap year or not.
Test Data : 2016
Expected Output :
2016 is a leap year.

5. Write a C# Sharp program to read the age of a candidate and determine whether it is eligible for casting his/her own vote.
Test Data : 21
Expected Output:
Congratulation! You are eligible for casting your vote.

6. Write a C# Sharp program to read the value of an integer m and display the value of n is 1 when m is larger than 0, 0 when m is 0 and -1 when m is less than 0.
Test Data : -5
Expected Output:
The value of n = -1

7. Write a C# Sharp program to accept a person's height in centimeters and categorize them according to their height.
Test Data : 135
Expected Output :
The person is Dwarf.

8. Write a C# Sharp program to find the largest of three numbers.
Test Data :
Input the 1st number :25
Input the 2nd number :63
Input the 3rd number :10
Expected Output :
The 2nd Number is the greatest among three

9. Write a C# Sharp program to accept a coordinate point in an XY coordinate system and determine in which quadrant the coordinate point lies.
Test Data :
Input the value for X coordinate :7
Input the value for Y coordinate :9
Expected Output :
The coordinate point (7,9) lies in the First quadrant.

10. Write a C# Sharp program to determine the eligibility for admission to a professional course based on the following criteria:
Marks in Maths >=65
Marks in Phy >=55
Marks in Chem>=50
Total in all three subject >=180
or
Total in Math and Subjects >=140
Test Data :
Input the marks obtained in Physics :65
Input the marks obtained in Chemistry :51
Input the marks obtained in Mathematics :72
Expected Output :
The candidate is eligible for admission.

11. Write a C# Sharp program to calculate the root of a quadratic equation.
Test Data :
Input the value of a : 1
Input the value of b : 5
Input the value of c : 7
Expected Output :
Root are imaginary;
No Solution.

12. Write a C# Sharp program to read roll no, name and marks of three subjects and calculate the total, percentage and division.
Test Data :
Input the Roll Number of the student :784
Input the Name of the Student :James
Input the marks of Physics, Chemistry and Computer Application : 70 80 90
Expected Output :
Roll No : 784

Name of Student : James
Marks in Physics : 70
Marks in Chemistry : 80
Marks in Computer Application : 90
Total Marks = 240
Percentage = 80.00
Division = First

13. Write a C# Sharp program to read temperature in centigrade and display a suitable message according to the temperature state below:
Temp < 0 then Freezing weather
Temp 0-10 then Very Cold weather
Temp 10-20 then Cold weather
Temp 20-30 then Normal in Temp
Temp 30-40 then Its Hot
Temp >=40 then Its Very Hot
Test Data :
42
Expected Output :
Its very hot.

14. Write a C# Sharp program to check whether a triangle is Equilateral, Isosceles or Scalene.
Test Data :
50 50 60
Expected Output :
This is an isosceles triangle.

15. Write a C# Sharp program to check whether a triangle can be formed by the given angles value.
Test Data :
40 55 65
Expected Output :
The triangle is not valid.

16. Write a C# Sharp program to check whether an alphabet letter is a vowel or a consonant.
Test Data :
k
Expected Output :

The alphabet is a consonant.

17. Write a C# Sharp program to calculate profit and loss on a transaction.
Test Data :
500 700
Expected Output :
You can book your profit amount : 200

18. Write a C# Sharp program to calculate and print the electricity bill of a customer. From the keyboard, the customer's name, ID, and unit consumed should be taken and displayed along with the total amount due.
The charge are as follow :
Unit
Charge/unit
upto 199
@1.20
200 and above but less than 400
@1.50
400 and above but less than 600
@1.80
600 and above
@2.00

If bill exceeds Rs. 400 then a surcharge of 15% will be charged and the minimum bill should be of Rs. 100/-
Test Data :
1001
James
800
Expected Output :
Customer IDNO :1001
Customer Name :James
unit Consumed :800
Amount Charges @Rs. 2.00 per unit : 1600.00
Surchage Amount : 240.00
Net Amount Paid By the Customer : 1840.00

19. Write a program in C# Sharp to accept a grade and declare the equivalent description :
Grade

Description
E
Excellent
V
Very Good
G
Good
A
Average
F
Fail

Test Data :
Input the grade :a
Expected Output :
You have chosen : Average

20. Write a C# Sharp program to read any day number as an integer and display the name of the day as a word.
Test Data :
4
Expected Output :
Thursday

21. Write a program in C# Sharp to read any digit, display in the word.
Test Data :
4
Expected Output :
Four

22. Write C# Sharp program to read any Month Number in integer and display Month name.
Test Data :
4
Expected Output:
April

23. Write a program in C# Sharp to read any Month Number in integer and display the number of days for this month.
Test Data :

7
Expected Output:
Month have 31 days

24. Write a C# Sharp program that calculates the area of geometrical shapes using a menu-driven approach.
Test Data :
Input your choice : 1
Input radius of the circle : 5
Expected Output :
The area is : 78.500000

1. Write a program in C# Sharp to input a string and print it.
Test Data :
Input the string : Welcome, halalAlmashakl
Expected Output :
The string you entered is : Welcome, halalAlmashakl

2. Write a C# Sharp program to find the length of a string without using a library function.
Test Data :
Input the string : halalAlmashakl.com
Expected Output :
Length of the string is : 15

3. Write a C# Sharp program to separate individual characters from a string.

4. Write a program in C# Sharp to print individual characters of the string in reverse order.

5. Write a program in C# Sharp to count the total number of words in a string.
Test Data :
Input the string : This is halalAlmashakl.com

Expected Output :
Total number of words in the string is : 3

6. Write a program in C# Sharp to compare two strings without using a string library functions.
Test Data :
Input the 1st string : This is first string
Input the 2nd string : This is first string
Expected Output :
The length of both strings are equal and
also, both strings are equal.

7. Write a program in C# Sharp to count the number of alphabets, digits and special characters in a string.
Test Data :
Input the string : Welcome to halalAlmashakl.com
Expected Output :
Number of Alphabets in the string is : 21
Number of Digits in the string is : 1
Number of Special characters in the string is : 4

8. Write a program in C# Sharp to copy one string to another string.
Test Data :
Input the string : This is a string to be copied.
Expected Output :
The First string is : This is a string to be copied.

The Second string is : This is a string to be copied.

Number of characters copied : 31

9. Write a C# Sharp program to count the number of vowels or consonants in a string.
Test Data :
Input the string : Welcome to halalAlmashakl.com
Expected Output :
The total number of vowel in the string is : 9
The total number of consonant in the string is : 12

10. Write a C# Sharp program to find the maximum number of characters in a string.
Test Data :
Input the string : Welcome to halalAlmashakl.com
Expected Output :
The Highest frequency of character 'e'
appears number of times : 4

11. Write a C# Sharp program to sort a string array in ascending order.
Test Data :
Input the string : this is a string
Expected Output :
After sorting the string appears like :
a g h i i i n r s s s t t

12. Write a C# Sharp program to read a string through the keyboard and sort it using bubble sort.
Test Data :
Input number of strings :3
Input 3 strings below :
abcd
zxcv
mnop
Expected Output :
After sorting the array appears like :
abcd
mnop
zxcv

13. Write a program in C# Sharp to extract a substring from a given string without using the library function.
Test Data :
Input the string : This is a test string
Input the position to start extraction :5
Input the length of substring :5
Expected Output :
The substring retrieve from the string is : is a

14. Write a C# Sharp program to check whether a given substring is present in the given string.
Test Data :
Input the string : This is a Test String
Input the substring to search : Test
Expected Output :
The substring exists in the string

15. Write a C# Sharp program to read a sentence and replace lowercase characters with uppercase and vice-versa.
Test Data :
Input the string : This is a string
Expected Output :
After conversion, the string is : tHIS IS A STRING

16. Write a program in C# Sharp to check the username and password.
Test Data :
Input a username: uesr
Input a password: pass
Input a username: abcd
Input a password: 1234
Expected Output :
Password entered successfully!

17. Write a program in C# Sharp to search for the position of a substring within a string.
Test Data :
Input a String: this is a string
Input a substring to be found in the string: is
Expected Output :
Found 'is' in 'this is a string' at position 2

18. Write a C# Sharp program to check whether a character is an alphabet and not and if so, check for the case.
Test Data :
Input a character: Z

Expected Output :
The character is uppercase.

19. Write a program in C# Sharp to find the number of times a substring appears in a given string.
Test Data :
Input the original string : this is original string
Input the string to be searched for : str
Expected Output :
The string 'str' occurs 1 times

20. Write a program in C# Sharp to insert a substring before the first occurrence of a string.
Test Data :
Input the original string : this is a string
Input the string to be searched for : a
Input the string to be inserted : test
Expected Output :
The modified string is : this is test a string

21. Write a C# Sharp program to compare (less than, greater than, equal to) two substrings.
Expected Output :
str1 = 'computer', str2 = 'system'
Substring 'mp' in 'computer' is less than substring 'sy' in 'system'.

22. Write a C# Sharp program to compare two substrings that only differ in case. The first comparison ignores case and the second comparison considers case.
Expected Output :
str1 = 'COMPUTER', str2 = 'computer'
Ignore case:
Substring 'MP' in 'COMPUTER' is equal to substring 'mp' in 'compu
Honor case:
Substring 'MP' in 'COMPUTER' is greater than substring 'mp' in 'computer'.

23. Write a C# Sharp program to compare two substrings using different cultures and ignore the substring case.
Expected Output :
str1 = 'COMPUTER', str2 = 'computer'
Ignore case, Turkish culture:
Substring 'UT' in 'COMPUTER' is equal to substring 'ut' in 'computer'.

Ignore case, invariant culture:
Substring 'UT' in 'COMPUTER' is equal to substring 'ut' in 'computer'.

24. Write a C# Sharp program to compare the last names of two people. It then lists them in alphabetical order.
Expected Output :
Sorted alphabetically by last name:
Michel Jhonson
John Peterson

25. Write a C# Sharp program to compare four sets of words by using each member of the string comparison enumeration. The comparisons use the conventions of the English (United States) and Sami (Upper Sweden) cultures.
Note : The strings "encyclopedia" and "encyclopedia" are considered equivalent in the en-US culture but not in the Sami (Northern Sweden) culture.
Expected Output :
case = Case (CurrentCulture): False
case = Case (CurrentCultureIgnoreCase): True
case = Case (InvariantCulture): False
case = Case (InvariantCultureIgnoreCase): True
case = Case (Ordinal): False
case = Case (OrdinalIgnoreCase): True
.
1. Write a program in C# Sharp to display the first 10 natural numbers.
Expected Output :
1 2 3 4 5 6 7 8 9 10

2. Write a C# Sharp program to find the sum of the first 10 natural numbers.
Expected Output :
The first 10 natural number is :
1 2 3 4 5 6 7 8 9 10
The Sum is : 55

3. Write a C# Sharp program that displays the sum of n natural numbers.
Test Data : 7
Expected Output :
The first 7 natural number is :

1 2 3 4 5 6 7
The Sum of Natural Number upto 7 terms : 28

4. Write a C# Sharp program to read 10 numbers and find their average and sum.
Test Data :
Input the 10 numbers :
Number-1 :2
...
Number-10 :2
Expected Output :
The sum of 10 no is : 51
The Average is : 5.100000

5. Write a C# Sharp program to display the cube of an integer up to given number.
Test Data :
Input number of terms : 5
Expected Output :
Number is : 1 and cube of the 1 is :1
Number is : 2 and cube of the 2 is :8
Number is : 3 and cube of the 3 is :27
Number is : 4 and cube of the 4 is :64
Number is : 5 and cube of the 5 is :125

6. Write a program in C# Sharp to display the multiplication table of a given integer.
Test Data :
Input the number (Table to be calculated) : 15
Expected Output :
15 X 1 = 15
...
...
15 X 10 = 150

7. Write a program in C# Sharp to display the multiplication table vertically from 1 to n.
Test Data :
Input upto the table number starting from 1 : 8
Expected Output :

Multiplication table from 1 to 8
1x1 = 1, 2x1 = 2, 3x1 = 3, 4x1 = 4, 5x1 = 5, 6x1 = 6, 7x1 = 7, 8x1 = 8
...
1x10 = 10, 2x10 = 20, 3x10 = 30, 4x10 = 40, 5x10 = 50, 6x10 = 60, 7x10 = 70, 8x10 = 80

8. Write a C# Sharp program to display the n terms of odd natural numbers and their sums.
Test Data
Input number of terms : 10
Expected Output :
The odd numbers are :1 3 5 7 9 11 13 15 17 19
The Sum of odd Natural Number upto 10 terms : 100

9. Write a program in C# Sharp to display a right angle triangle with an asterisk.
The pattern like :

```
*
**
***
****
```

10. Write a program in C# Sharp to display a pattern like a right angle triangle with a number.
The pattern like :

```
1
12
123
1234
```

11. Write a program in C# Sharp to make such a pattern like a right angle triangle with a number which repeats a number in a row.
The pattern like :

```
1
22
333
```

4444

12. Write a C# Sharp program to make such a pattern like a right angle triangle with the number increased by 1.

The pattern like :

```
1
2 3
4 5 6
7 8 9 10
```

13. Write a program in C# Sharp to make such a pattern like a pyramid with numbers increasing by 1.

```
  1
  2 3
 4 5 6
7 8 9 10
```

14. Write a program in C# Sharp to make such a pattern like a pyramid with an asterisk.

```
  *
  * *
 * * *
* * * *
```

15. Write a C# Sharp program to calculate the factorial of a given number.

Test Data :

Input the number : 5

Expected Output :

The Factorial of 5 is: 120

16. Write a program in C# Sharp to display the n terms of even natural number and their sum.

Test Data :
Input number of terms : 5
Expected Output :
The even numbers are :2 4 6 8 10
The Sum of even Natural Number upto 5 terms : 30

17. Write a program in C# Sharp to make such a pattern like a pyramid with a number which will repeat the number in the same row.

```
   1
  2 2
 3 3 3
4 4 4 4
```

18. Write a program in C# Sharp to find the sum of the series [1-X^2/2!+X^4/4!- ....].
Test Data :
Input the Value of x :2
Input the number of terms : 5
Expected Output :
the sum = -0.415873
Number of terms = 5
value of x = 2.000000

19. Write a program in C# Sharp to display the n terms of harmonic series and their sum.
1 + 1/2 + 1/3 + 1/4 + 1/5 ... 1/n terms
Test Data :
Input the number of terms : 5
Expected Output :
1/1 + 1/2 + 1/3 + 1/4 + 1/5 +
Sum of Series upto 5 terms : 2.283334

20. Write a program in C# Sharp to display the pattern like pyramid using an asterisk and each row contain an odd number of an asterisks.

```
  *
 ***
*****
```

21. Write a program in C# Sharp to display the sum of the series [9 + 99 + 999 + 9999 ...].
Test Data :
Input the number or terms :5
Expected Output :
9 99 999 9999 99999
The sum of the series = 111105

22. Write a program in C# Sharp to print Floyd's Triangle.
1
01
101
0101
10101

23. Write a program in C# Sharp to display the sum of the series [1+x+x^2/2!+x^3/3!+....].
Test Data :
Input the value of x :3
Input number of terms : 5
Expected Output :
The sum is : 16.375000
Number of terms = 5
The value of x = 3.000000

24. Write a program in C# Sharp to find the sum of the series [x - x^3 + x^5 - x^7 + x^9 -.....].
Test Data :
Input the value of x : 2
Input number of terms : 5
The sum = 410
Number of terms = 5
The value of x = 2

25. Write a C# Sharp program that displays the n terms of square natural numbers and their sum.

1 4 9 16 ... n Terms
Test Data :
Input the number of terms : 5
Expected Output :
The square natural upto 5 terms are :1 4 9 16 25
The Sum of Square Natural Number upto 5 terms = 55